

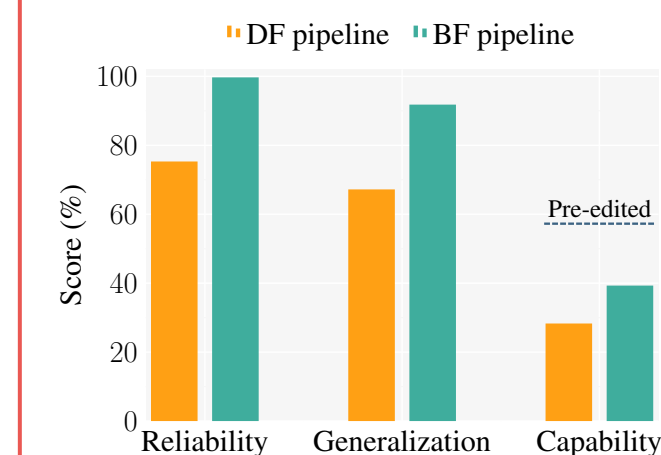
Fine-tuning Done *Right* in Model Editing

Wanli Yang, Rui Tang, Hongyu Zang, Du Su, Qi Cao, Jingang Wang, Huawei Shen, Xueqi Cheng, Fei Sun

How to Do Fine-tuning Right?

1

Pipeline

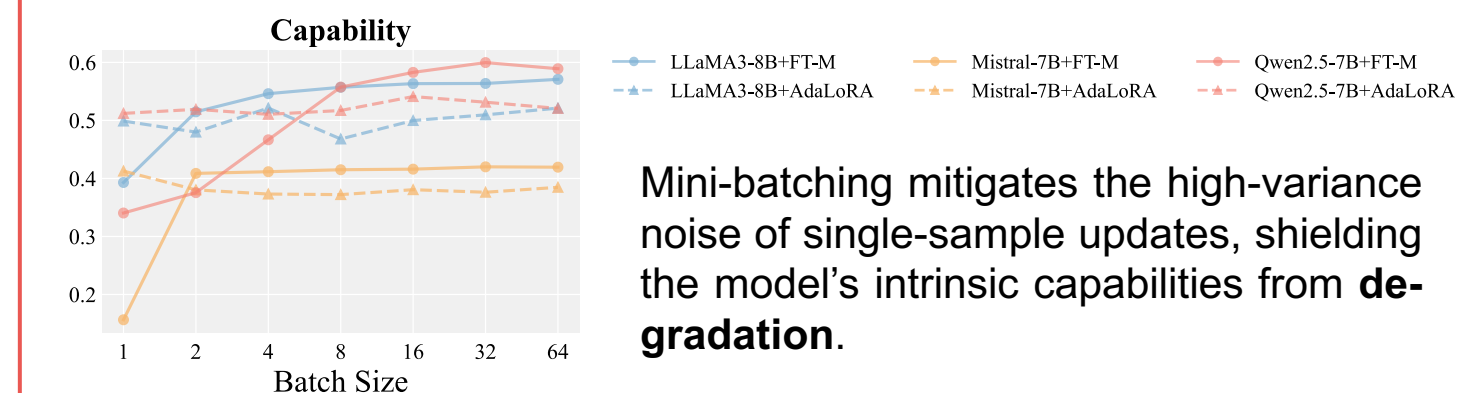


Method	ZsRE					
	Reliability		Generalization		Capability	
	DF	BF	DF	BF	DF	BF
LLaMA						
FT-L	0.00	0.00	0.00	0.10	14.96	23.47
FT-M	75.30	99.70	67.20	91.80	28.30	39.30
AdaLoRA	3.80	90.50	3.30	69.70	44.81	49.89
RoseLoRA	0.30	1.00	0.00	0.70	57.13	57.38

Conventional Depth-First editing leads to **Catastrophic Forgetting**. Our Breadth-First pipeline enables joint convergence, preserving all edits simultaneously.

2

Granularity

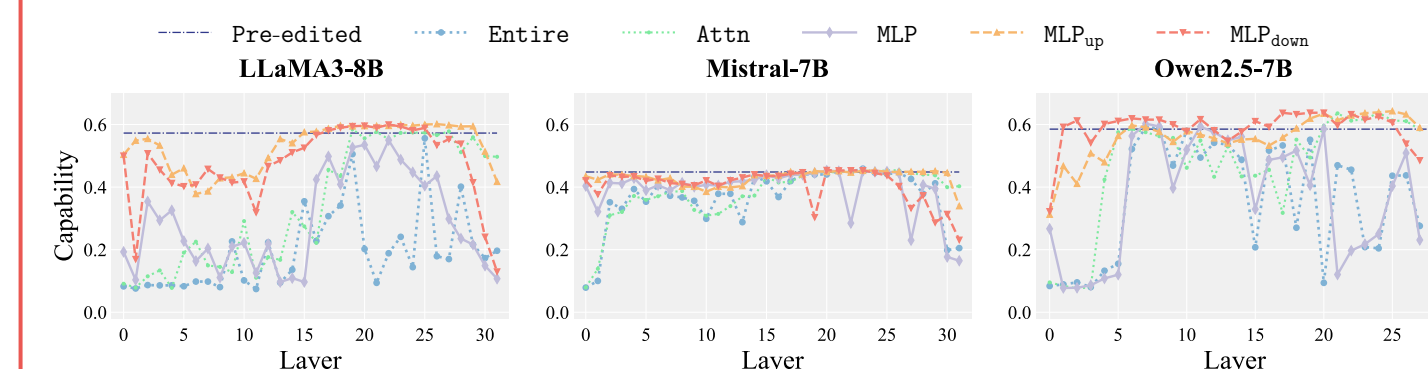


Mini-batching mitigates the high-variance noise of single-sample updates, shielding the model's intrinsic capabilities from **de-gradation**.

Where to Fine-tune?

3

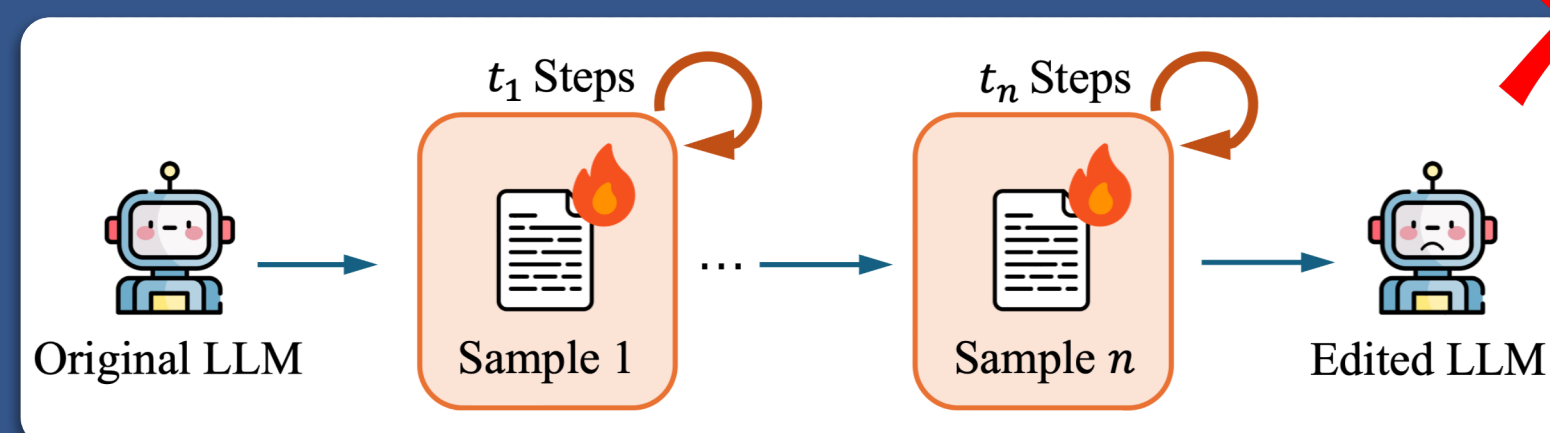
Location



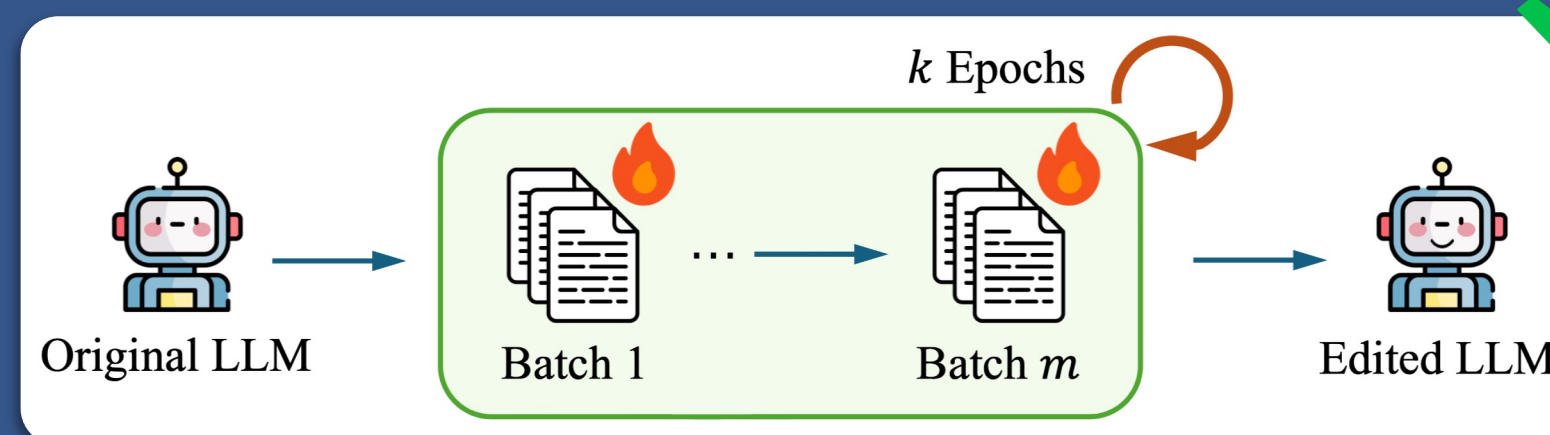
Editing success is **broadly achievable**: most locations reach near-perfect reliability. Location affects **side effects**: MLPdown in mid-to-late layers best preserves capability.

Fine-tuning is Not *Weak* in Model Editing

Wrong Pipeline, Wrong Conclusion



Depth-First Editing Pipeline



Breadth-First Standard Pipeline



Paper



GitHub



Outperform SOTA?

Yes!

Data	Method	LLaMA3-8B				Mistral-7B				Qwen2.5-7B			
		Rel.	Gen.	Cap.	Time	Rel.	Gen.	Cap.	Time	Rel.	Gen.	Cap.	Time
ZsRE	Pre-edited	—	—	57.26	—	—	—	44.82	—	—	—	58.52	—
	MEMIT	26.23	23.30	25.67	9.90	24.30	19.60	18.11	10.28	39.57	32.10	47.87	9.93
	RECT	0.03	0.03	14.98	25.38	0.17	0.27	14.88	25.79	9.70	8.20	16.06	24.31
	WISE	4.17	3.50	—	14.42	15.87	11.33	—	13.02	7.67	5.23	—	30.96
	AlphaEdit	64.50	40.57	54.71	12.31	3.30	3.00	15.13	10.95	2.70	2.33	16.31	10.55
	RLEdit	66.67	59.40	57.41	0.58	26.10	19.53	21.30	0.47	54.57	47.60	60.74	0.65
	UltraEdit	34.93	22.67	55.93	0.22	40.43	23.40	44.22	0.04	40.10	19.77	59.25	0.27
	LocFT-BF	98.97	75.83	57.04	0.27	98.93	49.57	42.73	0.31	98.87	74.37	59.07	0.49



Best reliability in all settings: **+33.7%** over the second-best.



Strong capability retention: **≤ 0.33%** loss across all settings.



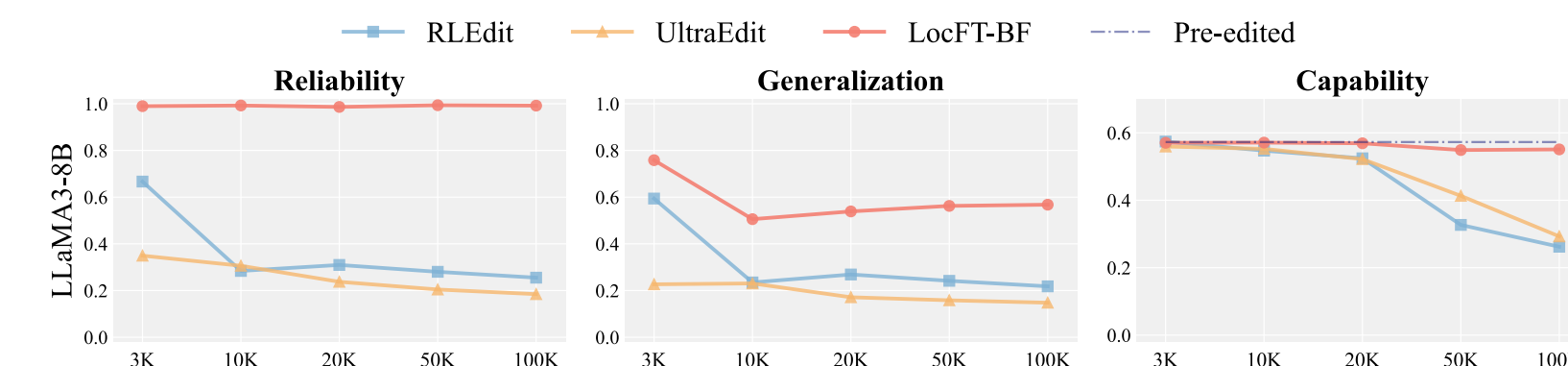
Top-tier editing efficiency: **50×** faster than locate-then-edit.

Scale to Real-world Editing?

Yes!



Stay stable to **100K Edits**



Seamlessly extended to **72B Models**

Strategy	Qwen2.5-7B			Qwen2.5-14B			Qwen2.5-32B			Qwen2.5-72B		
	Rel.	Gen.	Cap.	Rel.	Gen.	Cap.	Rel.	Gen.	Cap.	Rel.	Gen.	Cap.
Pre-edited	—	—	58.52	—	—	62.56	—	—	63.72	—	—	64.84
Default Pos.	99.20	83.40	59.33	99.00	78.80	61.62	99.30	70.00	60.27	98.60	65.30	63.64
Proportional Pos.	99.20	83.40	59.33	99.50	77.90	61.12	99.40	68.20	64.72	99.30	68.30	63.87